1.01 Proof				
1.01a 1.01d 1.01b 1.01c	Proof	a) Understand and be able to use the structure of mathematical proof, proceeding from given assumptions through a series of logical steps to a conclusion. *In particular, learners should use methods of proof including proof by deduction and proof by exhaustion.* b) Understand and be able to use the logical connectives $\equiv$, $\Rightarrow$, $\Leftrightarrow$. *Learners should be familiar with the language associated with the logical connectives: "congruence", "if..... then" and "if and only if" (or "iff").* c) Be able to show disproof by counter example. *Learners should understand that this means that, given a statement of the form "if $P(x)$ is true then $Q(x)$ is true", finding a single x for which $P(x)$ is true but $Q(x)$ is false is to offer a disproof by counter example.* *Questions requiring proof will be set on content with which the learner is expected to be familiar e.g. through study of GCSE (9–1) or AS Level Mathematics.* *Learners are expected to understand and be able to use terms such as "integer", "real", "rational" and "irrational".*	d) Understand and be able to use proof by contradiction. *In particular, learners should understand a proof of the irrationality of $\sqrt{2}$ and the infinity of primes.* *Questions requiring proof by contradiction will be set on content with which the learner is expected to be familiar e.g. through study of GCSE (9–1), AS or A Level Mathematics.*	MA1

OCR Ref.	Subject Content	Stage 1 learners should ...	Stage 2 learners additionally should ...	DfE Ref.